

Mechanistic Failure Mode Investigation and Resolution of Parvovirus Retentive Filters

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Filter C is highly likely to be the VPro based on a publication by LaCasse in 2013 on the same topic where VPro was named. Filter A is highly likely to be the Planova 20N based on various publications on the impact of pressure change

PP7 = 26 nm = phi174

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Virus retentive filters are a key product safety measure for biopharmaceuticals. A simplistic perception is that they function solely based on a size-based particle removal mechanism of mechanical sieving and retention of particles based on their hydrodynamic size. Recent observations have revealed a more nuanced picture, indicating that changes in viral particle retention can result from process pressure and/or flow interruptions. In this study, a mechanistic investigation was performed to help identify a potential mechanism leading to the reported reduced particle retention in small virus filters. Permeate flow rate or permeate driving force were varied and analyzed for their impact on particle retention in three commercially available small virus retentive filters. © 2016 American Institute of Chemical Engineers Biotechnol. Prog., 000:000–000, 2016

Keywords: monoclonal antibody, purification, virus retentive filtration, depth filtration, viral clearance, parvovirus, bacteriophage, convective flow, in membrane diffusion

Introduction

Small virus retentive filtration (VRF) is a key unit operation in the comprehensive viral clearance package for biopharmaceutical and plasma-derived products.^{1,2} VRF is generally considered to be a robust unit operation; small virus retentive filters are expected to provide greater than 4 log₁₀ clearance of parvovirus-sized (or larger) virus particles and near complete clearance of larger viruses of concern like retroviruses. This performance expectation has allowed the development of nomenclature standards for both large and small virus retentive filters based on bacteriophage based removal.³ However, like most technologies, small virus

retentive filter performance has gaps and failure modes when operated under extreme, non-recommended conditions. Previously identified gaps are extreme flux decay⁴ and virus overloading.⁵ Recently, fluid flow interruptions (FFI) have been identified as a technical vulnerability for more than one type of small virus retentive filters.^{6–8} Recent work has also established that the magnitude of low pressure or FFI impacts on small virus retention are both product and process-specific.^{6,9–11} A phenomena we targeted understanding with these studies was “in-membrane particle diffusion” within the tortuous matrix. We speculate within the membrane matrix at the nano-scale particles could diffuse perhaps 1–15 μm within the matrix to access a larger diameter pore during a diffusive favorable flow regime which may enable particles to permeate through the membrane following a long duration FFI or very low convective flow rates. Process FFI

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are common in large scale manufacturing operations that use bags or tanks to transfer the process intermediate to and from the virus filter. If the process intermediate is stored in multiple bags, FFIs would occur as bags are disconnected and reconnected to the feed pump during processing or valves are manipulated. Further, many large scale operations include a recovery buffer wash step at the end of filtration to maximize yield. This step would temporarily interrupt fluid flow resulting in a pressure reduction while buffer is transferred to the load tank or when the process plumbing configuration is manipulated. To increase product recovery, the recovery buffer wash material is typically collected in the same pool as the VRF process filtrate. Although interruptions may be avoided by using specialty valves or system configurations that may allow seamless transfer between the multiple process intermediate loads and buffer, this is not always possible in existing manufacturing facilities. Finally, process or FFIs can be the consequence of unexpected events such as power outage, equipment failure or by activation of an emergency stop of the system.

A recent study observed reduced parvovirus retention following a FFI or under low VRF flowrate conditions for select small virus retentive filters.¹² Additional studies have confirmed the general observation and extended it to other filter types; in which small virus retention performance can be vulnerable to changes in flow or pressure, albeit in a process and product specific manner.^{6–8} These observations were made mostly with mammalian parvovirus models Minute Mouse Virus (MMV) and Porcine Parvovirus (PPV) as virus models.

Bacteriophage have been employed in the past to evaluate filter performance^{4,5,13–16} and to classify filter types (i.e., large and small virus retentive filters³). Bacteriophage have been utilized for these studies [e.g., PP7 (Pseudomonas phage), Phi-X174 (Enterobacteria phage)] because they are similar in shape while being slightly larger than parvovirus,^{14,17} although they have yet to be used for regulatory claims. However, the rapid growth kinetics of bacteriophage and their ease of use allow for large scale, high-throughput studies necessary to establish process robustness, define retention mechanisms, and fully evaluate failure modes of small virus retentive filters. In addition, their hardness and icosahedral geometry provides assurance that the filter study measures virus sieving, not mechanical stress on the spiked particle as a clearance mode.

In these studies, we used high titers of the bacteriophage PP7 to probe performance under stress conditions and mechanisms of failure for three commercially available small virus retentive filters, Filter A, Filter B, and Filter C. For confirmation we then repeated a subset of the conditions with titers of MMV typical of small scale validation studies as a mammalian parvovirus model.

Materials and Methods

Viruses and assays

Bacteriophage PP7 was prepared and titer measured as per procedures recommended in the PDA technical report 41: Virus Retentive Filtration.³ Previous studies with small bacteriophage¹⁴ had determined that a spiking titer of 10^7 – 10^8 PFU/mL is optimal to allow a large window for LRV measurement, without overloading or impacting the flow of most

virus filters. Spiking for the studies in this manuscript targeted at 5×10^8 PFU/mL to help increase detection of particle retention differences on these larger phage particles while understanding differences of the three filters evaluated. The extremely high titers of bacteriophage employed in these studies were orders of magnitude above the likely titers of parvovirus contamination that would reach the filtration stage in biopharmaceutical processes, and somewhat higher than would typically targeted for a scale down validation study. However, the high titers allowed a large enough LRV window to enable precise measurements of changes in particle passage for the larger PP7 particles after a flow interruption under the various conditions we tested. Thus, to evaluate mechanisms, the titer needed to be higher compared with a typical parvovirus MMV spike used to generate a regulatory claim for a small virus filter.

MMV, Crawford strain, was obtained from the American Type Cell Culture (ATCC) and propagated in 324K cells. To minimize filter fouling due to non-virus associated components, WuXi AppTec's Ultra-3 grade MMV preparations were used for this study.

For each trial, PP7 was spiked into the VRF feedstream to a target titer of 5×10^8 PFU/mL, while MMV targeted a spike level of 5×10^6 PFU/mL, after gentle swirling to mix, spiked feeds were sampled, and loaded to their appropriate VRF system. Samples from these spiked process fluid feeds were held at room temperature for the duration of the tests then stored at 2°C–8°C prior to assay for titer along with filtrate samples generated during the experiments.

MMV titers in load and filtrate samples were determined using an MMV plaque assay using 324K indicator cells (WUXi AppTec Procedure 35189). Where low counts were expected (filtrate samples), large volume plating techniques were also used. Log₁₀ reduction values (LRV) were calculated by determining the log₁₀ of the quotient, total virus in the spiked feed divided by the total virus in the filtrate as displayed in Equation (1) below.

Equation (1): Calculating the log-reduction value of a virus filter

$$\text{LRV} = \log_{10}[(C_{fd} \times V_{fd}) / (CF \times VF)] \quad (1)$$

where, C_{fd} is the concentration of virus in the feed; V_{fd} is the volume of virus spiked feed processed through the filter; CF is the concentration of virus in the filtrate; and VF is the volume of the filtrate.

Feedstreams and feedstream conditioning

To maintain data consistency, a single sub-aliquoted process feedstream was used as a filtration feedstock in this study (10 g/L mAb solution in 25 mM acetate, 175 mM NaCl, pH 5.2). The feedstream was a typical VRF load source, a process intermediate from a recombinant (Chinese Hamster Ovary Cell-derived) biotherapeutic process. In particular the biotherapeutic was a monoclonal antibody, subclass IgG1 with a calculated isoelectric point of 8.9 and a molecular weight of 144 kDa. Prior to freezing and storage, the process intermediate was prefiltered using a Millistak+[®] X0HC depth filtration device (EMD Millipore Catalog no. MX0HC027H1, 270 cm²). The prefiltration device was flushed with WFI and equilibrated with buffer prior to feedstream filtration following vendor recommendations. The Millistak+[®] X0HC was then used to filter the feedstream

Table 1. Summary of VRF Tests, Samples, Filters, and Spike Particle and Conclusion for Each Study

Study Description	Filtrate Samples	Virus Filter(s)	Virus/Bacteriophage	Conclusion
Multistep pressure reduction (MSPR-PP7)	P1, P2, P3, P4, P5, P1 After 10-min FFI	Filters A, B, C	PP7	All filters displayed some decreased phage retention to different degrees with reducing pressure or FFI. Filters B and C showed robust phage retention (LRV >4) in all cases.
Multistep pressure reduction (MSPR-MMV)	P1, P2, P3, P4, P5, P1 After 10-min FFI	Filters A, C	MMV	Filter A displayed decreased MMV retention with reducing pressure or FFI
Rapid pressure reduction (RPR-PP7)	P1, P5, P1 After 10-min FFI	Filters A, B, C	PP7	All filters displayed decreased phage retention to different degrees. Filters B and C showed robust phage retention (LRV >4) with extended low pressure and following FFI
Rapid pressure reduction (RPR-MMV)	P1, 5× cycle P1-0, P1 After 10-min FFI	Filters A, C	MMV	Filter A and C displayed decreased MMV retention to varying degrees after 10 min FFI
High feed pressure, low trans-membrane pressure (HFP-LTMP-PP7)	High TMP, Low TMP	Filters A, B, C	PP7	Filters A and B displayed decreased phage retention , (although still robust phage reduction of LRV > 4) at low TMP similar to P5 in MSPR and RPR
High feed pressure, low trans-membrane pressure (HFP-LTMP-MMV)	High TMP, Low TMP	Filters A, C	MMV	Filter A displayed decreased MMV retention at low TMP
High viscosity (HV-PP7)	P1 Before FFI, P1 After 10-min FFI	Filters A, B, C	PP7	Filter A and B displayed decreased phage retention (although still robust phage reduction of LRV > 4) following a 10 min FFI with 6 cP fluid viscosity. HV improved phage retention following FFI
FFI duration study (FFI)	P1 Before FFI, P1 After FFI	Filter A	MMV	A 1–3 min interruption had no significant impact on LRV but, a 10 min interruption resulted in decreased LRV

Slight decrease in retention for Filter C, still have robust LRV of 5.4, page 6 and Figure 4C

at a flux of 100 L/m²h for a total volumetric loading of 200 L/m². Inclusion of this prefiltration step was intended to reduce the levels of potential small virus retentive filter foulants like aggregates and impurities. With this treatment, our studies could evaluate differences in particle retention associated with FFI and operational pressures which control VRF process flow rate, without the risk of significant plugging or flow decay of the virus retentive filters.¹⁸ The prefiltered VRF feedstream was aliquoted into sterile 1 L Nalgene PETG bottles (Nalgene Part# 3110-42), flash frozen by submerging in liquid nitrogen and stored at −80°C.

The VRF feedstream was stored and shipped at −80°C to the two test sites (WuXi AppTec and CDER), the aliquots were thawed as required at 25°C for approximately 14 h before filtration experiments. Following thawing, the material was again filtered through a sterilizing grade 0.2 μm PES vacuum filter prior to spiking with PP7 or MMV.

To prepare a high viscosity variant of the feedstream a predetermined level of sucrose (450 g/L) was added in order to increase the solution viscosity approximately six-fold to 6 centipoise (cP) (temperature range of 18°C–25°C).¹⁹ Beet derived sucrose was mixed with the thawed feedstream on a magnetic mixing plate, operating at 360 rpm for 60 min to ensure full dissolution of the sucrose. The high viscosity feedstream was again filtered through a sterilizing grade 0.2 μm PES vacuum filter prior to spiking with PP7.

Filtration experiments

Three commercially available scaled-down small virus retentive filters were investigated in this study. The three filters identified as A, B, and C in this body of work included three different membrane chemistries each with different membrane morphologies.

The scale-down VRF devices were operated within the vendor specified operating pressure limits for all steps. The VRF devices were flushed and equilibrated with the buffer for 15–20 min at their upper target feed pressure until the vendor-recommended flushing or wetting volumetric throughput was exceeded. The VRF devices were challenged with spiked feedstream where filtrate fractions were collected as detailed in Table 1. Each trial filtrate fraction volumetric target was 20 ± 2 mL, representing 20–65 L/m² volumetric challenge across the three scale-down model devices used in these trials.

The predefined filtrate volume of 20 mL was chosen to represent greater than 10 times the operational liquid filtrate hold-up volume in the VRF devices to ensure each filtrate fraction was not significantly impacted via dilution from the device hold up volume of the prior condition. Between one and six filtrate fractions were collected for each VRF trial (see Table 1), based on the pre-specified experimental protocol. Each VRF filtrate fraction was collected into a separate sterile 50 mL conical tube (Corning Part# CLS430290) during the filtration trials.

All VRF trials were performed in duplicate; all the PP7 and MMV LRV results presented in each figure are the average of two duplicate parallel trials for each test condition. All presented duplicate runs demonstrated log reduction values within 1.0 log₁₀ to verify comparable and reproducible titer assay and experimental performance. All filtration trials were operated with constant feed pressure mode, with pressure set points varied between experimental conditions and filtrate fractions. The applied feed pressure was controlled with a constant pressure nitrogen gas overlay on the feedstock pressure reservoir, with the vendor recommended pressure for the filter type defined as “P1 pressure.” Applied gas overlay pressure was measured and controlled by Anderson

digital gauges (Anderson Part# 068P01000411) and a 0–60 pounds per square inch (psi) pressure regulator (IngerSol Rand part# PR4021-200).

Multistep pressure reduction (MSPR)

In these trials, pressure was reduced stepwise to determine the point where virus breakthrough may change. The filters were first equilibrated with buffer. Next, during the load processing stage, all devices started at upper feed pressure target defined by vendor recommendations for the P1 filtrate fraction as detailed in Figure 1A. No back pressure was applied on the filtrate side during all feedstream processing. Following the collection of 20 mL of filtrate, the pressure was reduced stepwise to four pressure set points that spanned the typical operational pressure range for each device: P2, P3, P4, and P5, where 20 mL of filtrate was collected at each set point. The filtrate fraction collection at P5 was followed by a 10-min FFI. To achieve this, the system feed pressure was reduced to zero psi followed by closing of the feed valves for the 10-min duration of the FFI. The feed pressure was returned to P1 set point for each filter device, and the feed valves were then opened and a 20 mL filtrate sample was collected, P1 after 10-min FFI fraction.

Rapid pressure reduction (RPR)

The filters were first equilibrated with buffer. Next, during the spiked feedstream processing stage, all devices started at the target upper feed pressure P1 for the duration of P1 filtrate fraction as detailed in Figure 3A with no back pressure applied on the filtrate side of the devices. Two studies were performed which differed in implementation: one with PP7 and one with MMV. During the PP7 tests the second filtrate fraction occurred at the lowest pressure P5, as detailed in Figure 4A. Following the P5 filtration sample collection, the filters were exposed to a 10-min FFI while system feed pressure reduced to zero psi and feed valve closed for the duration of the FFI. Following the 10-min FFI, the feed pressure was returned to the P1 set points and feed valves were opened for a final filtrate sample collection P1 after 10-min FFI. The MMV trials were configured slightly different with the goal of exploring how very rapid pressure cycling will impact virus breakthrough. The initial filtrate fraction was performed as described above for the PP7 trial. However, the second filtrate fraction was processed at P1 following a 5× pressure cycle where the feed pressure dropped to zero psig and back to P1 five times in a one minute time span by rapid venting of the feed vessel pressurized gas. Following the 5× pressure cycle filtration sample collection, the trial filters were exposed to a 10-min FFI with system feed pressure reduced to zero psig and the feed valves closed. Following the 10-min FFI, the feed pressure was returned to the P1 set points and feed valves were opened for a final filtrate sample collection P1 after 10-min FFI.

High feed pressure low transmembrane pressure (HFP-LTMP)

In these trials the feed pressures utilized were the high upper feed pressures P1, from the previous MSPR study. However, the pressure on the filtrate side of these devices was regulated by connecting to a second pressure vessel that was pressurized to 2 psig below the feed pressure vessel, resulting in a net trans-membrane pressure (TMP) of 2 psig.

The filtrate was collected into the pressure vessel on the filtrate side of the device. Because the transmembrane pressure was low, the operational duration was 2–3 h to deliver 20 mL of filtrate.

High viscosity (HV) feed, high feed pressure, high transmembrane pressure

The feedstream was modified in these trials, by adding a predetermined level of sucrose (450 g/L) in order to increase the solution viscosity approximately six-fold to 6 centipoise (cP).¹⁸ As before the filters were first equilibrated with buffer. During the spiked feedstream processing stage of the trials, all filter devices were operated at the same upper feed pressure, P1, with no back pressure present on the filtrate side during processing (Figure 7A). Following the initial feedstream filtration fraction, P1 before FFI collection, all trial filters were exposed to a 10-min FFI as described in the previous MSPR study. Following the 10-min FFI, the feed pressure was returned to P1 set point for each filter device for a final filtrate fraction collection P1 after 10-min FFI.

FFI duration study

As before the Filter A devices were first equilibrated with buffer. The load processing stage for filter devices started at the target upper feed P1 set point with no back pressure. Following the initial filtration fraction collection, all trials were exposed to a FFI of variable length, with valve configurations as described in the previous MSPR study. The duration of the FFI was varied to include 1, 3, and 10 min. Following the FFI, the feed pressure was returned to P1 set point for each filter device, and a final filtrate sample collection took place at P1 after FFI as described in the previous MSPR study. Table 1 summarizes the trials and samples for each study and virus filter.

Results and Discussion

Previous studies have demonstrated that more than one type of small virus retentive filters can be vulnerable to parvovirus passage if a pressure release or FFI is introduced during processing,^{6–8} albeit the magnitude has been demonstrated to be product and process specific.^{6,12–14} To confirm these observations while using our process fluids, and to determine if small bacteriophage (i.e., pseudomonas phage PP7) can serve as a model to study the mechanism behind this filter vulnerability, we compared passage of high titer spikes of PP7 and MMV before and after FFI with three filter types.

A primary goal of these studies was to understand the interplay of convective fluid flow and in-membrane particle diffusion relating to small particle retention performance of three commercial small virus retentive filters. The small virus retentive filters can be thought of as a heterogeneous tortuous matrix, which results in a pore size distribution, with some pores potentially being larger and less retentive to 20 nm virus particles. As we consider “in-membrane particle diffusion” within the tortuous matrix, we speculate it could be at the nano-scale where a particle could diffuse perhaps 1–15 μm within the matrix (not the entire membrane thickness) to access a larger diameter pore enabling it to permeate through the membrane under a diffusive favorable flow regime. Others have started to visualize the particle

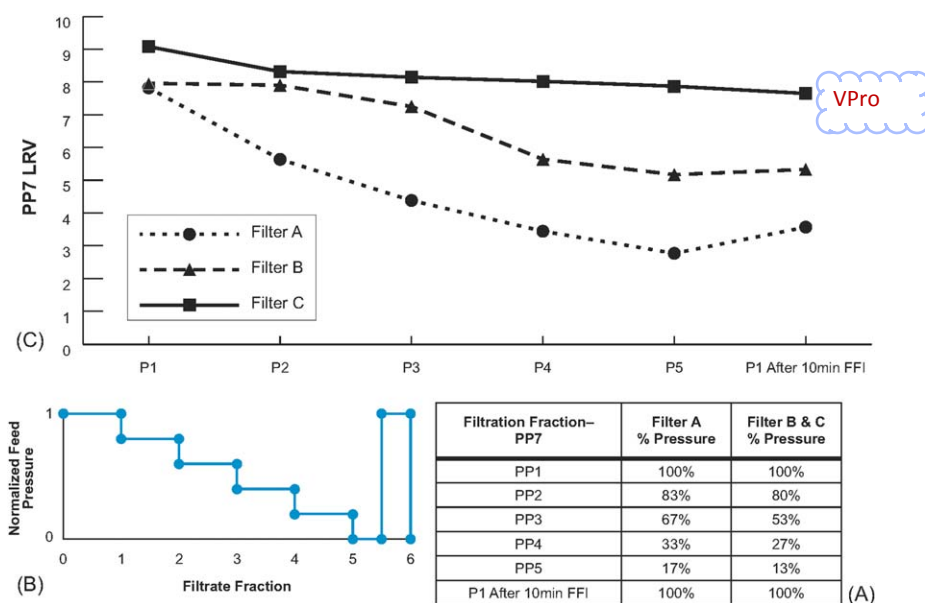


Figure 1. (A) VRF operational pressure in relation to the upper target pressure P1 that each filtration fraction was processed at. (B) A schematic of the normalized operational pressure the VRF devices were exposed to during the MSPR test for the various filtrate fractions. (C) Depicts MSPR-PP7 $\log_{10}$ reduction values vs. filtrate fraction, *open symbols* depict non-detectable phage particles in filtrate, *solid symbols* depict quantified particles in filtrate. As operational pressure was reduced all filters displayed reduced phage retention. All data points are the average of two duplicate runs for each condition with LRV results within 1.0 $\log_{10}$ of each other.

migration through select VRF membranes under similar low convective flow and favorable in-membrane diffusion operational conditions.⁹

An alternate hypothesis could be differential pressure related membrane compression changing the pore size or morphology of the filters resulting in altered retention performance. If the membranes were compressing or experiencing a morphological change related to transmembrane pressure, it would be highly likely to observe non-linear relation between water hydraulic permeability and pressure. Because the water hydraulic permeability vs. pressure of the three VRF products were all linear between 1 psi and the upper test pressures P1 in our studies, this scenario is not supported.

Multi-step pressure reduction

This study was designed to differentiate the interplay of convective fluid flow vs. in-membrane particle diffusion within the small virus retentive membrane matrix and their impact on small particle breakthrough. Our hypothesis was as VRF feed or differential pressure is reduced, the fluid convective flow is reduced. As a result, in-membrane particle diffusion from a tight pore to a more open pore within the tortuous matrix, could be more significant leading to higher levels of small particle breakthrough on small virus retentive filters. Multistep pressure reduction experiments were performed to determine the pressure point where PP7 and MMV breakthrough may change for each of the three filter types. The MSPR results with high titers of PP7 displayed a reduction in bacteriophage retention on Filters A, B, and C as operational pressure was reduced. Because very high titers of phage were employed, the LRV window was sufficiently large that some FFI related reduction in retention was observed in each filter type. Filter A was most impacted where retention decreased about 5 $\log_{10}$ over the pressure course, while Filter B retention notably decreased nearly 3 $\log_{10}$ and Filter C retention slightly diminished about 1.5

$\log_{10}$ (Figure 1A,C). The PP7 retention was the lowest for each of the three filters at P5 and P1 after 10-min FFI, indicating low operational pressure and long FFI may present the lowest small virus retention performance. Filter C in these PP7 trials displayed the highest retention of PP7 across the full pressure regime, with a slight drop as pressure was reduced. These observations were consistent with the hypothesis that multiple filter types can experience pressure and FFI-sensitive passage, *albeit* at different magnitudes. The operational feed pressures for each fraction in relation to P1 are displayed in Figure 1A, while Figure 1B depicts the normalized pressure profile during the MSPR VRF trials.

The MSPR results with MMV Figure 2C displayed a drop in small parvovirus retention for Filter A, observed at the P2 and lower pressure set points. The P2 set point represents 67% of the vendor recommended upper pressure, P1 target for Filter A as detailed in Figure 2A. In the subsequent filtrate fractions, there was a gradual drift downward in MMV retention for Filter A with reducing pressure. At pressure set points P4, P5, and P1 after 10-min FFI Filter A displayed a significant drop in MMV retention with an LRV of approximately 3.5 $\log_{10}$ compared with an LRV of 6.5 $\log_{10}$ at P1 the highest pressure set point. Indeed, evidence suggests MMV is slightly smaller in diameter than PP7.^{14,17} Thus, it is important to confirm bacteriophage observations with a smaller parvovirus model as it is used to make regulatory claims and may be a more sensitive indicator for breakthrough. For Filter C, there was no change in retention of MMV quantified across the full pressure regime explored; however, our LRV window using MMV was smaller than using PP7 and it is conceivable that with a significantly higher spiked titer of MMV breakthrough could be detected.

Rapid pressure reduction

The goal of this study was to further understand the relation between convective flow vs. in-membrane particle

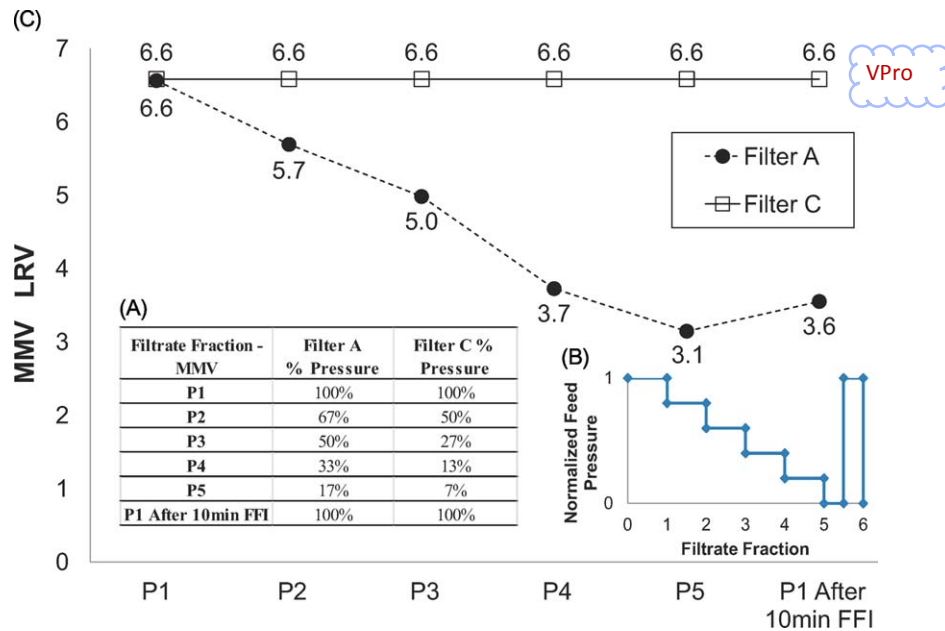


Figure 2. (A) VRF operational pressure in relation to the upper target pressure P1 that each filtration fraction was processed at. (B) A schematic of the normalized operational pressure the VRF devices were exposed to during the MSPR test for the various filtrate fractions. (C) Depicts MSPR-MMV Log₁₀ reduction values vs. filtrate fraction, *open symbols* depict non-detectable MMV particles in filtrate, *solid symbols* depict quantified particles in filtrate. As operational pressure was reduced Filter A displayed reduced MMV retention, **while Filter C did not**. All data points are the average of two duplicate runs for each condition with LRV results within 1.0 log₁₀ of each other.

diffusion influenced particle breakthrough, while looking at the pathway dependence of the particle breakthrough when compared with the sequential gradual pressure reduction in the previous MSPR studies. Our hypothesis was similar to the MSPR study, but by changing the pathway from high pressure to low pressure we would expect to observe similar levels of particle breakthrough at the lowest P5 condition regardless of the pathway to arrive at this condition. Similar particle breakthrough at low pressure or after a 10-min FFI may demonstrate the potential influence of in-membrane particle diffusion regardless of the previous process condition. In the MMV experiments the rapid pressure cycling was hypothesized to have little impact on changing the LRV if in-membrane diffusion was impacting the viral breakthrough, because there was very little time at the low pressure and diffusion is a time dependent phenomena. As can be seen in Figures 3C and 4C, at uninterrupted high feed pressures P1, all filters displayed the expected higher retention of both PP7 and MMV: ≥ 5.5 LRV measured for each filter under high convective flow conditions. For PP7, Filters A and B displayed a significant reduction in particle retention, where retention dropped about 3.0 LRV following a 10-min FFI compared with the retention when the typical high target operational pressure P1 is uninterrupted. **For MMV, Filters A displayed a significant reduction in particle retention, following a 10-min FFI compared with the retention when the typical high target operational pressure P1 is uninterrupted. Filter C displayed a slight reduction in retention, PP7 retention dropped 1.3 log₁₀, while MMV retention dropped 1.2 log₁₀ following the same 10-min FFI.** This is consistent with previous observations⁶ that virus passage post-FFI can occur with more than one filter type, *albeit* at different magnitudes.

For the MMV test condition, the second filtrate fraction 5× cycled P1-0 displayed viral retention performance within 1 log₁₀ of the P1 fraction, indicating that short term pressure change or fluctuation may not significantly diminish small

virus retention on the evaluated small virus filters Figure 4C. **In the Filter C 10-min FFI filtrate sample, the observed MMV LRV was 5.44, which is robust MMV retention although lower than the P1 or cycled pressure results.**

The comparable results of the two model viruses (MMV and PP7) indicate that bacteriophage, PP7 appears to be capable of serving as a good model for parvovirus for mechanistic studies of virus passage related to operational pressure, flux rates or following FFI events.

High feed pressure low TMP

This study was performed to investigate the effect of operational feed pressure and transmembrane pressure (TMP) targets on virus breakthrough, with back pressure placed on the permeate side of the filters. This study was intended to simulate a manufacturing process that may collect into a pressurized storage tank or contain hydraulic resistance from any plumbing or filtration downstream of the VRF. Our hypothesis was as TMP is reduced across the VRF convective flux through the membrane is reduced, where in-membrane particle diffusion could become more significant leading to higher particle breakthrough similar to the feed pressure P5 in the MSPR and RPR trials, regardless of the elevated inlet feed pressure in this study. The previous MSPR and RPR studies had demonstrated select small virus retentive filters had exhibited reduced MMV and PP7 retention as feed pressure and convective flux across the VRF was reduced. In the HFP-LTMP study, similar low convective flux to P5 in the MSPR study, while the filter devices were exposed to the high feed pressure P1, under a low differential pressure regime. Filters A and B displayed reduced retention performance under this high feed pressure and low TMP vs. the high TMP condition. The initial filtrate fractions in the low TMP trials displayed about 2–4 log₁₀ lower retention of PP7 and MMV compared with the high TMP

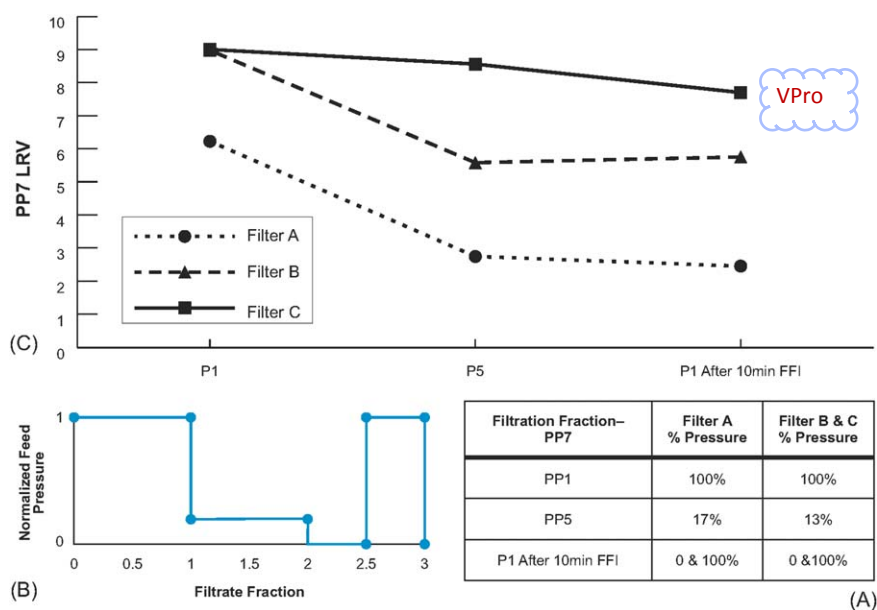


Figure 3. (A) VRF operational pressure in relation to the upper target pressure P1 that each filtration fraction was processed at. (B) A schematic of the normalized operational pressure the VRF devices were exposed to during the MSPR test for the various filtrate fractions. (C) Depicts RPR-PP7 Log_{10} reduction values vs. filtrate fraction, *open symbols* depict non-detectable phage particles in filtrate, *solid symbols* depict quantified particles in filtrate. As operational pressure was reduced multiple filter types displayed reduced phage retention to varying degrees. All data points are the average of two duplicate runs for each condition with LRV results within 1.0 log_{10} of each other.

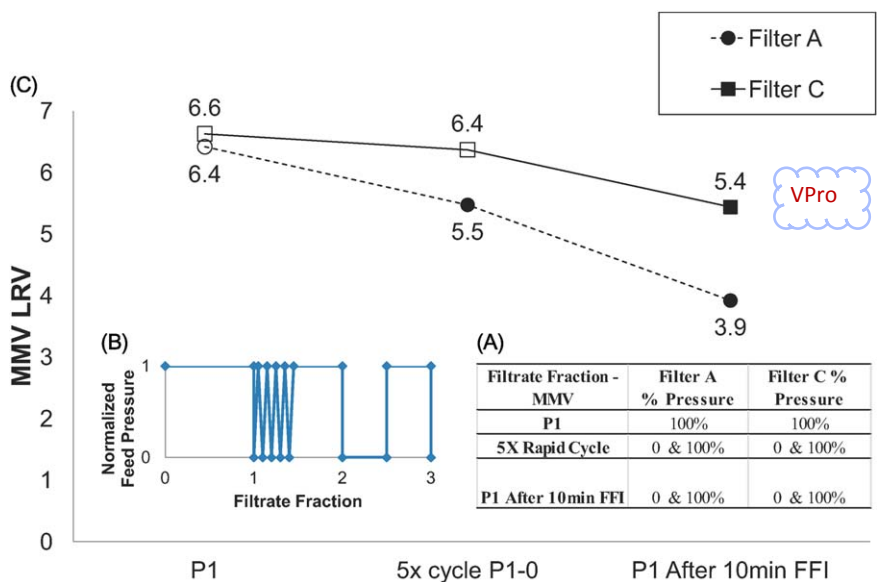


Figure 4. (A) VRF operational pressure in relation to the upper target pressure P1 that each filtration fraction was processed at. (B) A schematic of the normalized operational pressure the VRF devices were exposed to during the RPR test for the various filtrate fractions. (C) Depicts RPR-MMV Log_{10} reduction values vs. filtrate fraction, *open symbols* depict non-detectable MMV particles in filtrate, *solid symbols* depict quantified particles in filtrate. All data points are the average of two duplicate runs for each condition with LRV results within 1.0 log_{10} of each other.

condition, shown in Figures 5C and 6C. In the low TMP condition, Filter A displayed the lowest retention of PP7 (4.2 LRV) and MMV (3.3LRV), while Filter C displayed high retention of PP7 (8.7 LRV) and MMV (≥ 6.3 LRV) when challenged with the same PP7 or MMV load titer as detailed in Table 2. Filter B particle retention was intermediate in the low TMP PP7 testing (6.0 LRV), while not evaluated in the MMV trials.

These results indicate that the overall viral retention performance for Filters A and B using our process fluids can be

highly dependent on TMP which controls the process flux through the membrane. One could postulate the particle breakthrough is related to total volumetric throughput across the VRF. For example, in the MSPR study reduced particle retention was observed in the later filtrate fractions. Volumetric throughput appears secondary and less significant compared with process flux and/or operational pressures, since these initial 20 mL filtrate fractions demonstrated drastically lower PP7 and MMV retention at low TMP compared with the high TMP condition initial filtrate fractions. The

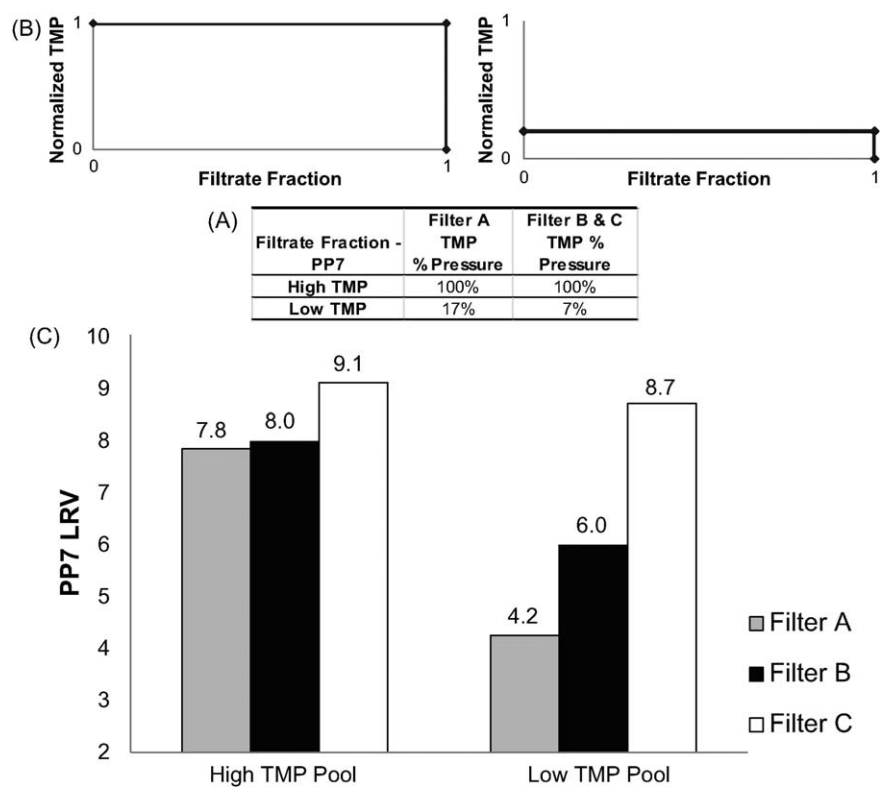


Figure 5. (A) VRF operational pressure in relation to the upper target pressure P1 that each filtration fraction was processed at. (B) Schematics of the normalized operational pressure the VRF devices were exposed to during the HFP-LTMP trials for the various filtrate fractions. (C) Depicts HFP-LTMP-PP7 Log₁₀ reduction values vs. filtrate fraction, $\geq$ symbols depict non-detectable PP7 particles in filtrate. All data points are the average of two duplicate runs for each condition with LRV results within 1.0 log₁₀ of each other.

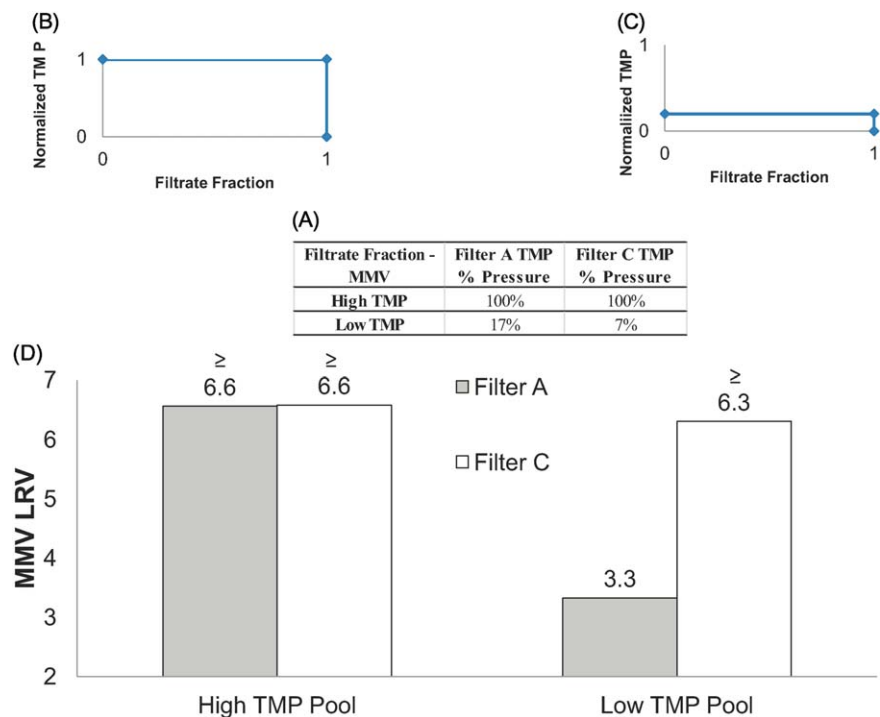


Figure 6. (A) Indicates the VRF operational pressure in relation to the upper target pressure P1 that each filtration fraction was processed at. (B, C) Schematics of the normalized operational pressure the VRF devices were exposed to during the HFP-LTMP trials for the various filtrate fractions. (D) Depicts HFP-LTMP-MMV Log₁₀ reduction values vs. filtrate fraction, $\geq$ symbols depict non-detectable MMV particles in filtrate. All data points are the average of two duplicate runs for each condition with LRV results within 1.0 log₁₀ of each other.

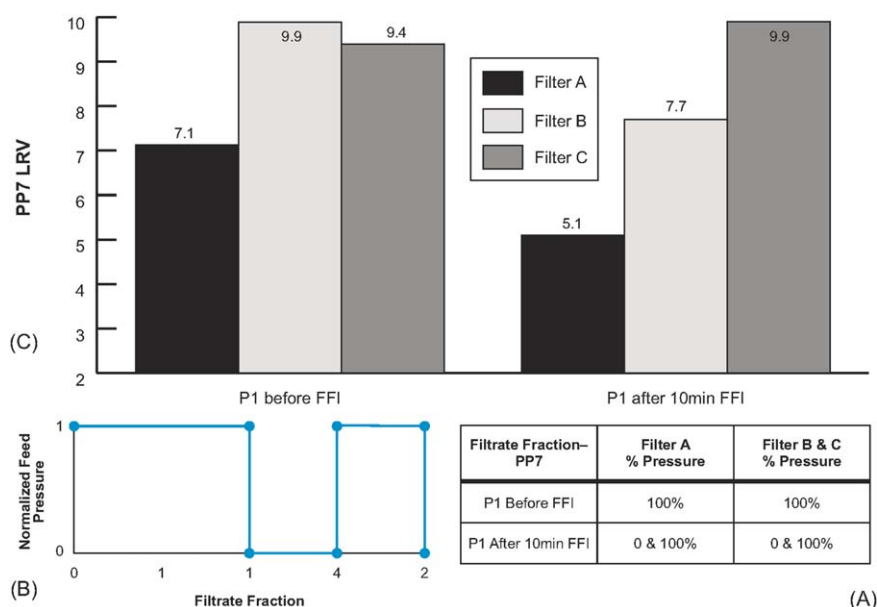


Figure 7. (A) Indicates the VRF operational pressure in relation to the upper target pressure P1 that each filtration fraction was processed at. (B) A schematic of the normalized operational pressure the VRF devices were exposed to during the high viscosity trials for the various filtrate fractions. (C) Depicts High Viscosity-PP7 Log₁₀ reduction values with and without FFI, $\geq$ symbols depict non-detectable PP7 particles in filtrate. All data points are the average of two duplicate runs for each condition with LRV results within 1.0 log₁₀ of each other.

HFP-LTMP tests condition simulated operation under low convective flow due to low differential pressure, despite high overall feed pressure. The observation that Filters A and B displayed reduced virus retention under this HFP-LTMP regime supports the hypothesis that in-membrane particle diffusion could be partially responsible for virus breakthrough behavior in small virus retentive filters when stress conditions favor it. Under the reduced convective flow environment in-membrane particle back diffusion could become more significant in the combined particle mass transfer environment while there is less forward convective flow to counteract the free particle diffusion, despite similar fluid characteristics at high feed pressure.

Here again in the HFP-LTMP studies, PP7 bacteriophage appears to be capable of serving as a good stress-model for parvovirus breakthrough studies on small virus retentive filters and for evaluating mechanisms of breakthrough modes.

High viscosity experiments

To further understand the hypothesis of in-membrane diffusional dependence on particle breakthrough, the load viscosity was increased in these experiments which would reduce the rate of diffusion and process flux. At high viscosity while operating at low convective flow, higher particle retention may support the hypothesis that in-membrane particle diffusion could be contributing to particle breakthrough on some small virus retentive filters. To achieve low process flux while operating at high TMP, the model VRF feedstream was modified to include 450 g/L of sucrose, an artificial fluid composition which is much more viscous than a typical bioprocess intermediate. It is important to note that as viscosity increases both membrane feedstream permeability and particle diffusion are hindered. In practice, the VRF process flux for each of the filters in the high viscosity test was very close to the flux at the P5 condition in the MSPR study as well as the HFP-LTMP test condition due to the

six-fold reduction in fluid permeability from the added viscosity. Here with the process fluid viscosity increased six-fold to about 6 cP, the particle diffusion rate inside the filter matrix and the liquid flow rate would decrease about six-fold compared with the similar 1 cP process fluid. As can be seen in Figure 7C, the PP7 particle retention by Filter A and Filter B after 10-min FFI was reduced about 2 log₁₀ compared with prior the FFI, while Filter C displayed no change in PP7 retention under these conditions. The data also suggests that the increase in viscosity helped to reduce in magnitude the decreased particle retention of the three filters evaluated, even following a 10-min FFI. Altogether this indicates that particle retention may be increased in a high viscosity environment where particle diffusion is hindered assuming there are no bacteriophage aggregates formed in the presence of sucrose. The fact that significant bacteriophage breakthrough did not occur before the FFI in these high viscosity trials argues that particle passage was not solely related to low VRF flux. Further studies at viscosities between 1 and 6 cP with additional FFI durations are required to better understand the relationship between convective flow, in-membrane particle diffusion and virus retention as related to fluid viscosity.

Overall the high viscosity results further support the hypothesis that in-membrane particle diffusion may be a factor influencing particle passage or retention in small virus retentive filters. Although operation of small virus removal filters with higher viscosity fluids may help stabilize robust viral clearance, this is likely not feasible in a manufacturing setting because the process time would be too long or the required membrane area would have to increase significantly.

FFI duration study

This study was designed to understand diffusional time dependence vs. membrane morphology impact of pressure changes. If the VRF membrane was compressed or

Table 2. Summary of VRF Tests, Filters, Load Particle Titer, and LRV Range for Each Study

Study Acronym	Small Virus Retentive Filter	PP7 Load Titer at Time of Assay (PFU/mL)	VRF Observed PP7 LRV ($\log_{10}$)	MMV Load Titer at Time of Assay (PFU/mL)	VRF Observed MMV LRV ($\log_{10}$)
MSPR	Filter A	7.30E+08	3.5–7.8	7.20E+06	3.1–6.6
	Filter B	9.50E+08	5.3–8.0	N/T	N/T
	Filter C	2.50E+09	7.6–9.1	7.60E+06	≥ 6.6
Rapid pressure reduction	Filter A	7.20E+08	2.5–6.2	5.33E+06	3.9 to ≥ 6.4
	Filter B	9.50E+08	5.7–9.0	N/T	N/T
	Filter C	2.50E+09	7.7–9.0	6.13E+06	5.4 to ≥ 6.6
Transmembrane pressure	Filter A HighTMP	7.30E+08	7.8	7.20E+06	≥ 6.6
	Filter B HighTMP	9.50E+08	8.0	N/T	N/T
	Filter C HighTMP	2.50E+09	9.1	7.60E+06	≥ 6.6
	Filter A LowTMP	9.50E+08	4.2	7.60E+06	3.3
	Filter B LowTMP	9.50E+08	6.0	NA	NA
	Filter C LowTMP	7.20E+08	8.7	7.60E+06	≥ 6.3
High viscosity	Filter A	7.70E+08	5.1–7.1	N/T	N/T
	Filter B	7.70E+08	7.7 to ≥ 9.9	N/T	N/T
	Filter C	7.70E+08	9.4 to ≥ 9.9	N/T	N/T
Pause duration	1 min PD	N/T	N/T	5.90E+06	4.8–5.4
	3 min PD	N/T	N/T	5.70E+06	5.1–5.7
	10 min PD	N/T	N/T	7.20E+06	3.6–6.6

PP7 titer assay limit of quantitation equals 1 CFU/mL.
 MMV titer assay limit of quantitation equals 2 CFU/mL.
 N/T, condition not tested.

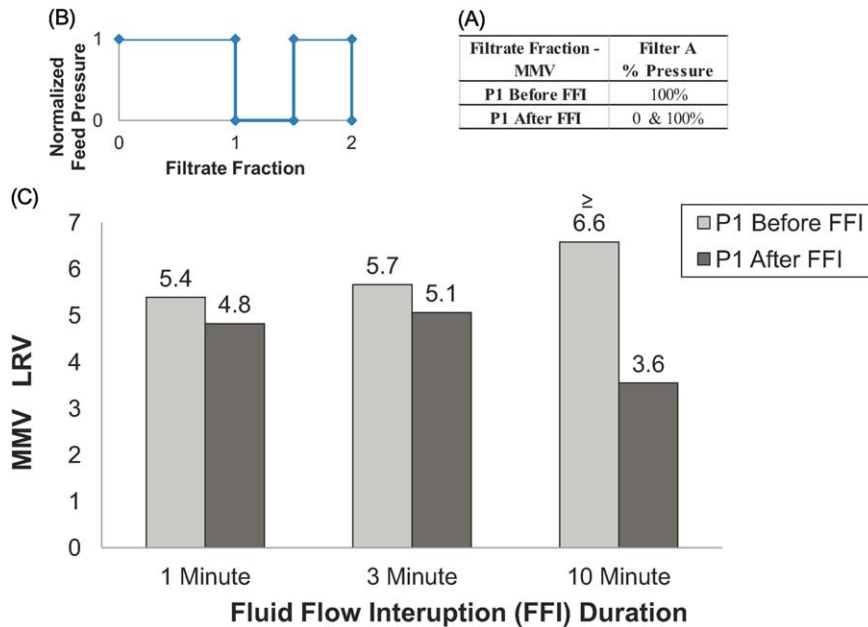


Figure 8. (A) Indicates the VRF operational pressure in relation to the upper target pressure P1 that each filtration fraction was processed at. (B) A schematic of the normalized operational pressure the VRF devices were exposed to during the FFI trials for the various filtrate fractions. (C) Depicts the FFI duration–Filter A: MMV $\log_{10}$ reduction values vs. filtrate fraction, $\geq$ symbols depict non-detectable MMV particles in filtrate. All data points are the average of two duplicate runs for each condition with LRV results within 1.0 $\log_{10}$ of each other.

experienced elastic deformation under high transmembrane pressure theoretically improving particle retention, we would expect the impact of any duration FFI on observed particle retention should be similar. Because most polymers have high modulus of elasticity their recoils rates are very short, so we speculate the mechanical equilibration to be achieved within a few seconds following a reduction in TMP. However if in-membrane particle diffusion were contributing to particle breakthrough it would be logical to speculate higher rates of particle breakthrough with increasing duration of a FFI, since a particle could back diffuse further with more time. This study was designed to determine if there was a time dependence impact of the FFI on MMV breakthrough

for Filter A, as this one seemed to be the most sensitive to FFI in our other studies. The study would also provide insight around how long the FFI needs to exist before significant change in particle retention may occur. The FFI duration results indicate Figure 8C, at high feed pressures Filter A displayed the expected high retention of MMV with ≥ 5.3 LRV measured for each filter in the P1 before FFI filtrate fraction. Filter A displayed a significant loss in MMV retention following a 10-min FFI. The 1- and 3-min FFI durations impacted filter performance to a much smaller degree where the retention of MMV results were within 1 $\log_{10}$ before and after the FFI. These results indicate that particle retention under FFI conditions may be time-dependent. This further

supports our hypothesis that virus passage could be influenced by in-membrane particle diffusion in select small virus retentive filters. Further studies looking at more FFI time durations could enable better understanding of the time dependence for a given small virus retentive membrane.

Bioprocess Implications and Risk Mitigation Strategies. Biotherapeutic processes that include virus removal filters should include consideration of the risks associated with FFI, if they are part of the process or may happen during deviations. While we and others have observed that the magnitude of FFI associated breakthrough may vary between filter types and process fluids, some level of breakthrough was observed both here and by others for more than one filter type. Mitigation steps may include additional process controls or specific design elements in small scale viral clearance studies to mitigate the risk of reduced small virus clearance. Potential in-process control strategies may include tight control around upper and lower operational pressures or volumetric fluid flux to ensure consistent small virus removal. Process FFIs or extended pauses should be avoided, strictly controlled for frequency and duration or well understood to avoid compromising viral clearance capabilities. The mitigation strategy can consider the degree of FFI associated breakthrough expected based on filter type and/or process fluid.

Viral clearance demonstration trials should demonstrate the worst-case process conditions including FFIs that the manufacturing process, if they are built-in to the process or could happen in a deviation. In the case of our process fluid, the worst-case scenario for Filters A and B identified in this manuscript appears to be the lowest process differential pressure and longest FFI, individually or in combination, although this may differ for other process fluids. For instance, if a manufacturing process has two unavoidable, built-in FFIs in order to switch load tanks and to switch to buffer wash to improve recovery, the viral clearance demonstration trials should include at least two FFIs that meet or exceed the respective time duration in the manufacturing process. Another strategy could be to evaluate ways to eliminate the pauses, for example using special valves. An additional example pertains to a manufacturing process with operational pressure control range of 30–45 psig, and the range is within vendor recommendations. In this example, the viral clearance demonstrations should be performed at or below the lowest pressure of 30 psig, which is predicted to provide the lowest observed retention of a small virus particle in a filter that exhibits reduced particle retention with low differential or feed pressure.

As stated above, the magnitude of the FFI effect can vary with filter type, and Filter C in this manuscript appears to carry some, but a lesser, risk in terms of magnitude of changes in small virus retention with varying operational pressure or FFIs for this process stream. As a result, it could be argued that fewer risk mitigation controls may be required for our process fluid. This manuscript indicates the process control limits for pressure, flow rate and FFI with this filter could be much wider without negatively impacting the small virus clearance capacity to the same degree as filter A, but some consideration of risk should be taken. With small virus filters, it appears very important to understand how the operational pressure or flow ranges along with FFIs may impact the demonstrated small virus clearance of the process.

Conclusions

Lower virus filter transmembrane pressures (<70% of vendor recommended maximum upper pressure) and significant FFIs (>3 min) using typical low viscosity (about 1 cP) process fluids yielded a reduction in small mammalian parvovirus or small bacteriophage retention across multiple filter types to varying degrees. In filters with more pronounced reduction in retention, it is possible to mitigate these effects to some extent, through in-process control strategies such as tight control around upper and lower operational pressures or volumetric fluid flux and excursion duration to ensure consistent small virus removal. The finding that different small virus retentive filters are dissimilarly susceptible to small particle breakthrough argues that each filter and process requires individualized risk assessment, process understanding and control. It is speculated that in-membrane particle diffusion under prolonged reduced or stopped convective fluid flow may contribute to particle liberation in the filter matrix leading to reduced particle retention across small virus filters.

Notation

CDER = center for drug evaluation and research (food and drug administration)

FFI = fluid flow interruption: a stop or pause to the process filtrate flux and differential pressure across the virus removal filter

HFP-LTMP = high feed pressure, low trans-membrane pressure

HV = high viscosity: a modified VRF load stream with viscosity of 6 cP, to investigate the impact of increased viscosity on small particle retention in this manuscript

In-membrane diffusion = diffusion of a particle inside of a tortuous membrane matrix potentially enabling particles to diffuse from a tight retentive pore to a more open pore that would increase the probability of the particle permeating through the membrane matrix when convective fluid flow is present

LRV = log₁₀ reduction of virus, the removal rate of virus particles across an individual unit operation. Calculated by determining the log₁₀ of the quotient, total virus in the spiked feed divided by the total virus in the filtrate as displayed in Equation (1)

MMV = minute mouse virus, an 18–24 nm mammalian parvovirus species frequently used as a small virus model to access Biotherapeutic process viral clearance capabilities due to its small size and high resistance to chemical inactivation methods

MSPR = multi-step pressure reduction

PES = polyether sulfone, a common polymer that several membranes used in the biopharmaceutical industry are derived from

PETG = glycol-modified polyethylene terephthalate, is a copolyester that is a clear amorphous thermoplastic that has a wide range of thermal stability, which is leveraged for process intermediate storage bottles stored at temperatures as low as –80°C

PP7 = *Pseudomonas* phage

RPR = rapid pressure reduction

Small Virus = virus removal filters characterized by the ability to typically retain more than 3 log₁₀ of parvovirus sized particles

Trans-membrane = the pressure difference between the feed side pressure (TMP) of the membrane and filtrate side of the membrane. This represents the primary driving force for fluid to flow through a small virus removal filter

WFI = water for injection

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Literature Cited

- Chen D. Viral clearance using traditional, well-understood unit operations (session i): virus-retentive filtration. *PDA J Pharm Sci Technol.* 2014;68:38–50.
- Miesegaes G, Lute S, Aranaha H, Brorson K. Virus retentive filters. In: M. Flickenger, editor. *Encyclopedia of Industrial Biotechnology*. Hoboken, NJ: John Wiley & Sons, Inc; 2010.
- Parenteral Drug Association Virus Filter Task Force. Technical report no. 41, Virus Filtration. *PDA J Pharm Sci Technol.* 2008; 62:S4.
- Bolton G, Cabatingan M, Rubino M, Lute S, Brorson K, Bailey M. Normal flow virus filtration – detection and assessment of endpoint in bioprocessing. *Biotechnol Appl Biochem.* 2005;42: 133–142.
- Lute S, Bailey M, Combs J, Sukumar M, Brorson K. Virus passage after extended processing in small virus retentive filters. *Biotechnol Appl Biochem.* 2007;47:141–151.
- LaCasse D, Genest P, Pizzelli K, Greenhalgh P, Mullin L, Slocum A. The impact of process interruption on virus retention with small virus filters. *BioProcess Int.* 2013;11:34–44.
- Stuckey J, Strauss D, Venkiteswaran A, Gao J, Luo W, Quertinmont M, O'Donnell S, Chen D. A novel approach to achieving modular retrovirus clearance for a parvovirus filter. *Biotechnol Prog.* 2014;30:79–85.
- Zydney AL, Woods MA. Effects of a pressure release on virus retention with the Ultipor DV20 membrane. *Biotechnol Bioeng.* 2014;111:545–551.
- Adan-Kubo OJ. Virus behavior during filtration is affected by structure of filter membrane and Brownian motion of virus. Parenteral Drug Association Virus and TSE Safety Forum, Bethesda, Maryland, Presentation June 9–11, 2014.
- Hongo-Hirasaki T. Effect of antibody solution conditions on virus removal filter performance, Parenteral Drug Association Virus and TSE Safety Forum. Barcelona, Spain, Presentation June 28–30, 2011.
- Hongo-Hirasaki T. Process designing for virus removal filtration, Parenteral Drug Association Virus and TSE Safety Forum. Berlin, Germany, Presentation June 4–6, 2013.
- Asper M. Virus breakthrough after pressure release during virus retentive filtration, Parenteral Drug Association Virus and TSE Safety Forum. Barcelona, Spain, Presentation June 28–30, 2011.
- Aranha-Creado H, Oshima K, Jafari S, Howard G Jr, Brandwein H. Virus retention by a hydrophilic triple-layer PVDF microporous membrane filter. *PDA J Pharm Sci Technol.* 1997;51:119–124.
- Lute S, Riordan W, Pease LIII, Tsai D, Levy R, Haque M, Martin J, Moroe I, Sato T, Morgan M, Krishnan M, Campbell J, Genest P, Dolan S, Martin S, Meyer A, Zachariah M, Tarlov M, Etzel M, Brorson K. PDA virus filter task force, A consensus rating method for small virus-retentive filters. I. Method development. *PDA J Pharm Sci Technol.* 2008;62:318–333.
- Oshima KH, Evans-Strickfaden T, Highsmith A. Comparison of filtration properties of hepatitis B virus, hepatitis C virus and simian virus 40 using a polyvinylidene fluoride membrane filter. *Vox Sang.* 1998;75:181–188.
- Oshima KH, Evans-Strickfaden TT, Highsmith AK, Ades EW. The use of a microporous polyvinylidene fluoride (PVDF) membrane filter to separate contaminating viral particles from biologically important proteins. *Biologicals.* 1996;24:137–145.
- Slocum A, Burnham M, Genest P, Venkiteswaran A, Chen D, Hughes J. Impact of virus preparation quality on parvovirus filter performance. *Biotechnol Bioeng.* 2013;110:229–239.
- Genest P, Ruppach H, Geyer C, Asper M, Parrella J, Evans B, Slocum A. Artifacts of virus filter validation. *BioProcess Int.* 2013;11:54–61.
- Asadi M. 2007. Beet Sugar Handbook. Hoboken, NJ: John Wiley & Sons; Table A-12.

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